



LectureAssist: An Agentic, Multimodal AI Assistant for Lecture-Grounded Study Material

Iftikhar Ahmed · Salman Shafi · Jarinah Tasnim
Supervisor: Dr. Adnan Areefen

Problem

- Lecture knowledge is locked in **video, speech and board writing**.
- Generic quiz tools produce **ungrounded, off-topic** questions.
- Fixed difficulty ignores **each student's weak topics**.

Goal. Grounded, validated, adaptive study material — fully local, **zero API cost**, on one free Colab T4.

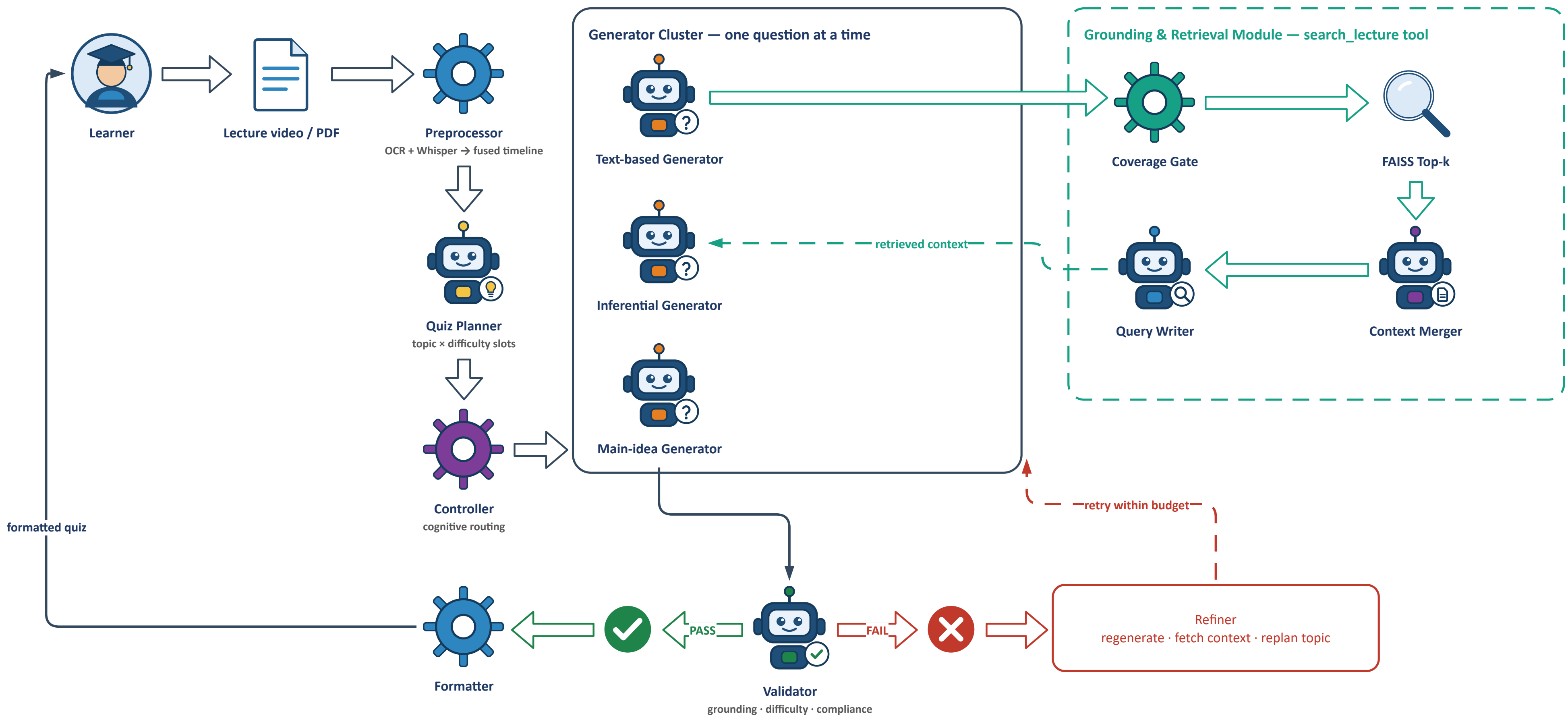
Evaluation Setup

- Source.** OpenStax *Calculus Vol. 1*, Limits chapter ($\approx 112k$ characters), also given to the judge as ground truth.
- Scale.** $n=200$ MCQ per pipeline — easy 68, medium 66, hard 66.
- Fairness.** Identical sliding window for both pipelines, so any difference reflects **prompting strategy alone**.
- Models.** Generator **gemma3:4b**, judge **gemma3:12b**, both local.

Stack

- Vision.** EasyOCR, per-surface preprocessing for slides, chalkboards and whiteboards.
- Speech.** Faster-Whisper with per-segment timestamps.
- Retrieval.** MiniLM embeddings in a FAISS cosine index.
- Serving.** Flask on Colab → ngrok → browser UI.

System Architecture



All agents run gemma3:4b served locally by Ollama; the independent judge is gemma3:12b. The Controller assigns each slot exactly one cognitive track, so the spread is enforced rather than requested.

Figure 1. Multimodal extraction feeds a grounded knowledge base; the orchestrator coordinates the specialist agents that produce every study artifact. Independent agents run concurrently and any agent crash is isolated.

Agentic Quiz Generation

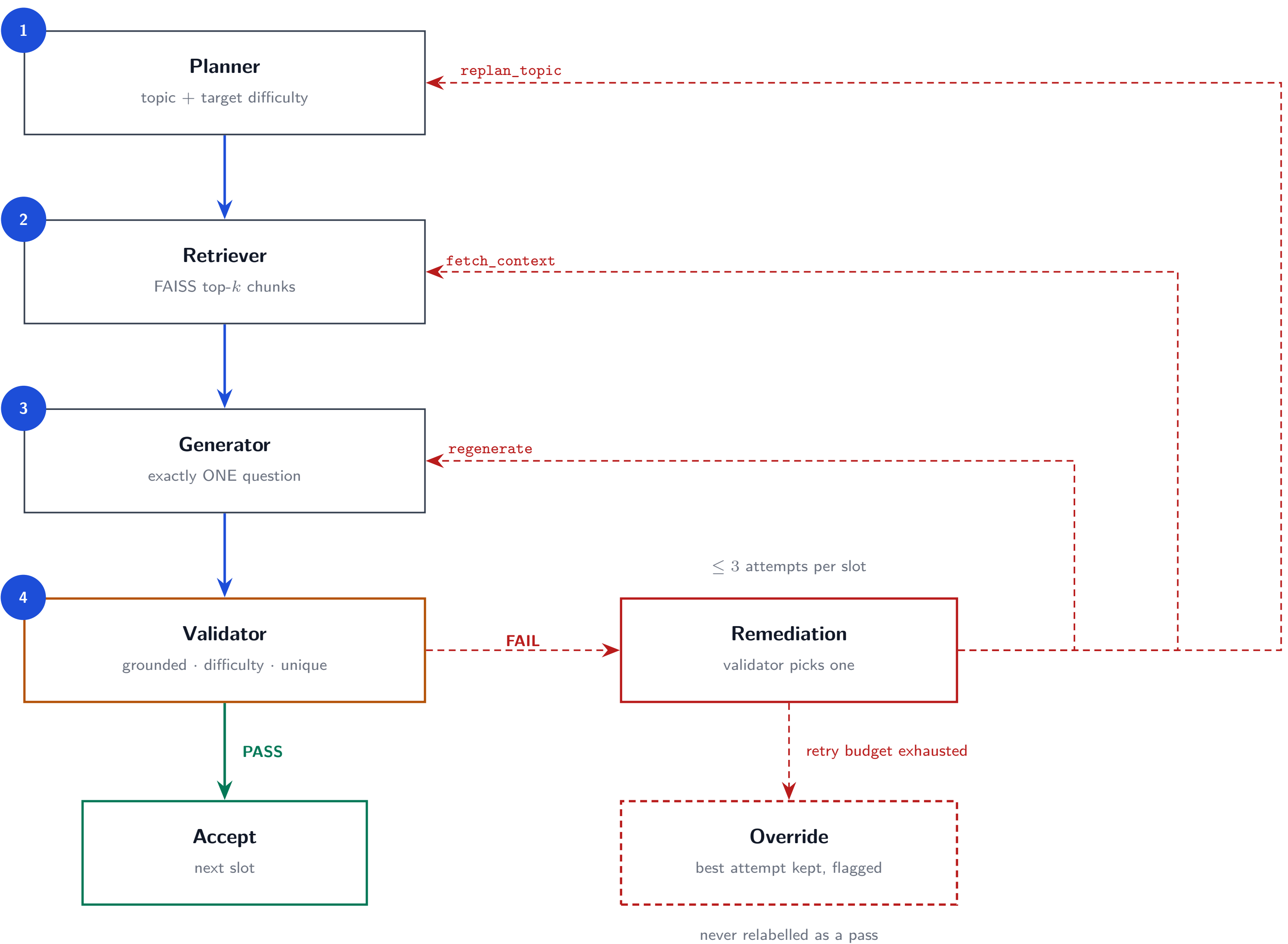


Figure 2. One question per slot; each must pass validation before the next begins. An exhausted retry budget flags an override, never a pass.

Results

Metric	Agen.	Non-ag.
<i>Judged 1–5 (saturated)</i>		
Overall	4.86	4.93
Correctness	4.98	5.00
Clarity	4.92	4.97
Difficulty	4.36	4.39
<i>Automated 0–1</i>		
Duplicate ↓	0.385	0.280
Grounding	0.557	0.624
Eff. accept	0.612	0.720
Answer match	0.940	0.970
Similarity ↓	0.932	0.894
Accept rate	0.995	1.000
Verified	0.995	0.995

Diff.	Pipe.	Dup.↓	Grnd.
Easy	Agen.	0.353	0.562
	Non-ag.	0.324	0.642
Medium	Agen.	0.379	0.595
	Non-ag.	0.227	0.614
Hard	Agen.	0.288	0.513
	Non-ag.	0.182	0.616

Diff.	Pipe.	Eff.	Match
Easy	Agen.	0.647	0.971
	Non-ag.	0.676	0.971
Medium	Agen.	0.612	0.879
	Non-ag.	0.773	0.955
Hard	Agen.	0.712	0.970
	Non-ag.	0.818	0.985

$n=200$ per pipeline. Green marks the better value; ↓ marks rows where lower is better. Per-difficulty $n=68/66/66$.

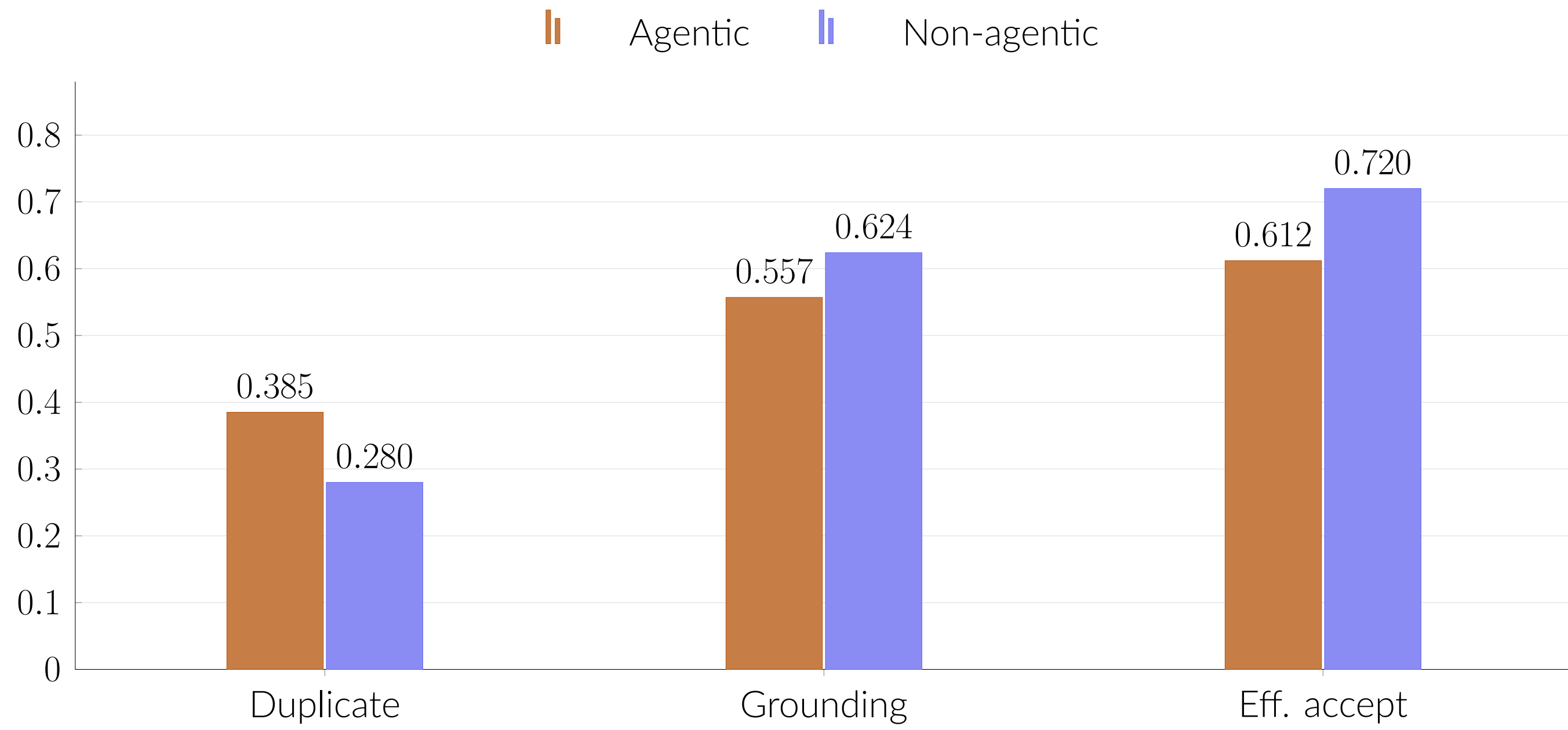


Figure 3. The three metrics that discriminate. Lower is better for duplicate rate, so the baseline leads on all three.

Conclusion

A fully local, \$0-API multimodal pipeline that turns a lecture into grounded, validated study material. The judged dimensions **saturate** (4.36–5.00), so only the automated metrics discriminate — and on those the agentic loop does **not** beat single-pass prompting. The engineering contribution stands; the scientific claim is **not established**, and we report it that way.